

NovaTech Executive Summary

This executive summary highlights three verified business insights identified using the NovaTech QuickSight dashboard and Amazon Q analysis. Inefficient marketing and post-sale product issues are hurting revenue by slowing down sales and risking customer retention; fixing these five connected areas together is key to growth.

Executive Snapshot

- Email Campaigns Have the Lowest Response Rate
- All Campaigns Are Generating Negative ROI
- Deal Loss Reasons Are Evenly Spread
- Sales Cycle Is Lengthening
- At-Risk Accounts Need Immediate Attention

Recommended Next Steps

1. Boost funnel volume without increasing spend.
2. A thorough audit of targeting, lead quality, and cost structure.
3. Multi-pronged approach: better qualification to filter poor-fit prospects, competitive positioning improvements, and more flexible pricing/timing options.
4. Streamline the sales process, improve handoff from marketing, and equip reps with better enablement tools.
5. Proactively address product pain points and prioritize outreach to at-risk accounts.

Key Insights

1. Email Campaigns Have the Lowest Response Rate
 - a. **Observation:** Email has the lowest response rate of 8.7%, behind Paid Social of 40.3%, and Direct Mail of 53.0%.
 - b. **Meaning:** Email is cheap and scalable - not optimizing resources to increase response rate to grow our sales pipeline for free.
 - c. **Action:** Fix email subject lines, target, and personalization.
2. All Campaigns Are Generating Negative ROI
 - a. **Observation:** Every marketing campaign is currently losing money ranging from NovaEdge Awareness at -\$267 to Q3 Growth Sprint at -\$381.
 - b. **Meaning:** Need to fix how we turn spend into revenue.
 - c. **Action:** Immediate audit of targeting, lead quality, and costs.
3. Deal Loss Reasons Are Evenly Spread
 - a. **Observation:** Losing deals to a mix of reasons led by Poor Product Fit at 43 count and Competitor Won at 34 count.
 - b. **Meaning:** Need a multi-pronged approach.
 - c. **Action:** Improve lead qualification, sharpen competitive positioning, and offer flexible pricing.
4. Sales Cycle Is Lengthening
 - a. **Observation:** Average time to close a deal has skyrocketed from 58 days in early 2024 to over 90 days in early 2025.
 - b. **Meaning:** Wasting resources and risking buyer abandonment.
 - c. **Action:** Streamline our workflows, fix marketing handoffs, and give reps better sales tools.
5. At-Risk Accounts Need Immediate Attention

- a. Observation: At-risk accounts are causing 30 support tickets tied to over \$53K in deal value, and nearly 23% of all tickets carry negative customer sentiment driven heavily by bugs in Notifications and the Analytics Dashboard.
- b. Meaning: Need to stop customer churn and protect revenue.
- c. Action: Run proactive outreach and fix these product areas.

Confidence Summary

Insight	Confidence
Email Campaigns Lowest Response Rate	High
All Campaigns Generating Negative ROI	High
Deal Loss Reasons Evenly Spread	High
Sales Cycle Lengthening	High
At-Risk Accounts Need Immediate Attention	High

Emerging Risks

- Burnout and resource waste from sluggish sales
- High churn risk and revenue loss post-sale
- Poor sales conversion due to a weak funnel mix

Data Sources Used

- CRM Deals CSV
- Support Tickets CSV
- Marketing Campaigns CSV
- QuickSight Interactive Dashboard
- Amazon Q Topic Analysis

Data Strategy

This dashboard draws on three separate data sources: a CRM system tracking 499 sales deals, a marketing platform tracking 2,240 leads and their campaign interactions, and a support system tracking 3,000 customer tickets. Before building anything, I ran a series of verification checks asking basic factual questions like row counts and known null values against each dataset and confirming the answers matched documented expectations. This caught data quality issues early, before they could quietly distort a chart or KPI later on.

Each dataset was imported into SPICE, QuickSight's in-memory storage engine that keeps large datasets fast to query without hitting the source files directly every time. From there, several fields needed type corrections for example, deal values and campaign spend needed to be stored as decimal numbers rather than whole numbers, since truncating cents would understate real dollar figures across hundreds of rows.

The three datasets share a common account identifier, but they don't share the same level of detail: the CRM table has exactly one row per deal, while a single account can appear across many marketing leads and many support tickets. Combining tables at different levels of detail is a common pitfall; directly combining them would have caused deal and revenue figures to be counted multiple times, once for every matching lead or ticket. To avoid this, I first collapsed the marketing and support data

down to one summary row per account (total leads, total tickets, recent ticket volume, and so on), and only then combined those summaries with the CRM data using a join a standard method of combining two tables based on a shared identifier, in this case the account ID. I used a left join, meaning every CRM account is kept in the result even where it happens to have no matching marketing or support activity. This produced a single unified dataset that correctly connects sales, marketing, and support activity at the account level, without inflating any of the underlying numbers.

On top of the corrected, joined data, I added several calculated fields new metrics derived from existing data using straightforward business logic, such as the number of days between a deal's creation and its close, or a flag identifying accounts showing signs of risk (high recent support volume, negative sentiment, and high deal value together).

Dashboard Design Rationale

The dashboard is organized into three sheets, each mapped directly to one of the three questions this project set out to answer: how effectively our marketing spend generates and converts leads, how our sales pipeline is performing, and how healthy our existing customer relationships are.

Each sheet (except Customer Health) opens with a KPI card covering the headline metrics a reader would want at a glance, such as overall win rate or total marketing spend. A KPI card was not included in Customer Health as the other visuals provide pertinent data and I wanted to limit visuals to what is absolutely necessary to provide the data and turn it into action. Below that, each chart type was chosen to match the shape of the underlying question rather than a one-size-fits-all approach: a funnel chart shows how leads narrow down through each stage from initial prospect to closed deal, since that's inherently a sequential process rather than a simple comparison; ranked bar charts answer "which of these is performing best or worst" questions, such as return on investment by campaign; and a scatter plot is used specifically for the at-risk account analysis, since that question is fundamentally about the relationship between two numbers (support ticket volume and deal value) rather than a single ranking.

The Customer Health sheet is the only one built partly on the unified, cross-source dataset described above, because it's the only view that genuinely needs to compare information from two different systems at once to answer whether our highest-maintenance accounts are also our most valuable ones. The Marketing Funnel and Sales Pipeline sheets are each built entirely from their own native data, since their questions don't require that cross-system view, and keeping them separate avoids unnecessary complexity.

Finally, the dashboard includes interactive filters so a viewer can narrow results by campaign, region, product, or similar attributes; a one-click filtering action so that selecting a bar in one chart automatically updates related charts on the same sheet; and a navigation link that lets a viewer click on a specific at-risk account and jump directly to that account's sales history on a different sheet. These interactions were included specifically because the original request emphasized wanting one connected view of an account across all three systems, rather than three disconnected reports.

Topic Configuration & Q Accuracy

Q agreed with the dashboard when I hadn't changed anything within the Campaign Channel Topic. I think setting Topics is more detailed work than we initially understand that can cause skewed insights if not done properly. Q would be used when in a hurry and the user wants to validate information without having to read a dashboard and/or without having to put together new queries into dashboard visuals.